Linear Programming

Linear Programming (LP)

- A model consisting of linear relationships representing a firm's objective and resource constraints.
- LP problem involves a linear objective functions, which is the function that must be maximized or minimized. This objective function is subject to some constraints, which are inequalities or equations that restrict the values of the variables.
- LP is used to find the best or optimal solution to a problem that requires a decision about how best to use a set of limited resources to achieve a goal or an objective.

Linear Programming Model

Max/Min $z = c_1x_1 + c_2x_2 + \ldots + c_nx_n$ → Objective Function

Subject to:

$$a_{11}x_1 + a_{12}x_2 + \ldots + a_{1n}x_n \ (\leq, =, \geq)\ b_1$$
$$a_{21}x_1 + a_{22}x_2 + \ldots + a_{2n}x_n \ (\leq, =, \geq)\ b_2$$
$$\vdots \qquad\qquad\qquad\qquad\qquad\qquad \vdots$$
$$\cdot \qquad\qquad\qquad\qquad\qquad\qquad \cdot$$
$$a_{m1}x_1 + a_{m2}x_2 + \ldots + a_{mn}x_n \ (\leq, =, \geq)\ b_m$$

Constraints

Where:

x_j = decision variables

b_i = constraint levels

c_j = objective function coefficients

a_{ij} = constraint coefficients

Approaches used in Linear Programming

1. **Geometric Approach**
   - graphical method
   - uses feasible region and corner points (coordinate points) to determine the optimal solution
2. **Algebraic Approach**
   - **SIMPLEX** method
   - developed by Goerge B. Dantzig in 1947
   - utilized if there are more decision variables and problem constraints

Example 1: Solve using **Graphical Method**.

Consider the LP model:

$$\text{Max } P = 30x + 40y$$

Subject to:

$$x + y \leq 5$$

$$2x + 3y \geq 6$$

$$x \geq 1$$

$$x \geq 0$$

$$y \geq 0$$

a. Shade the feasible region
b. Determine the corner points
c. Formulate decision

Example 2: Solve using **Graphical Method**.

Consider the LP model:

$$\text{Max } P = 4x + 5y$$

Subject to:

$$x + 2y \leq 10$$

$$6x + 6y \leq 36$$

$$x \leq 4$$

$$x, y \geq 0$$

a. Shade the feasible region
b. Determine the corner points
c. Formulate decision

Example 2: Solve using **Graphical Method**.

Consider the LP model:

$$\text{Minimize} \quad C = 5X + 2Y$$

$$\text{Subject to:} \quad X + 3Y \geq 12$$

$$X + Y \geq 8$$

$$X \geq 0$$

$$Y \geq 0$$

a. Shade the feasible region
b. Determine the corner points
c. Formulate decision

Application: Formulating LP Models

Step 1. Read the problem carefully. If appropriate, organize the data into a table.

Step 2. Determine and define the variables. These variables represent the unknown quantities whose values are what you want to find.

Step 3. Formulate the objective function. This is what you want to maximize or minimize.

Step 4. Formulate the constraints inequalities. These are expressions that limit the amount of resources you can use and the restrictions of your constraint variables.

Step 5. Solve the problem using an appropriate algorithm or mathematical technique manually and/or using any available mathematical software.

Example: Tent Production

A manufacturer of lightweight mountain tents makes a standard model and an expedition model for national distribution. Each standard tent requires 1 labor-hour from the cutting department and 3 labor hours from the assembly department. Each expedition tent requires 2 labor-hour from the cutting department and 4 labor hours from the assembly department. The maximum labor-hour available per day in the cutting department and the assembly department are 32 and 84, respectively. If the company makes a profit of P500 on each standard tent and P800 on each expedition tent, how many tents of each type should be manufactured each day to maximize the total daily profit?

Solution:

A manufacturer of lightweight mountain tents makes a standard model and an expedition model for national distribution. Each standard tent requires 1 labor-hour from the cutting department and 3 labor hours from the assembly department. Each expedition tent requires 2 labor-hour from the cutting department and 4 labor hours from the assembly department. The maximum labor-hour available per day in the cutting department and the assembly department are 32 and 84, respectively. If the company makes a profit of P500 on each standard tent and P800 on each expedition tent. How many tents of each type should be manufactured each day to **maximize the total daily profit** (Assuming all tents can be sold)?

Step 1. Read the problem carefully. If appropriate, organize the data into a table.

	Standard Model (Labor hrs/day)	Expedition Model (Labor hrs/day)	Maximum labor- hrs/ day
Cutting dept	1	2	32
Assembly dept	3	4	84
Profit/ tent	P500	P800	

Continuation…

Step 2. Determine and define the variables. These variables represent the unknown quantities whose values are what you want to find.

Let : x = number of standard tents produced per day

y = number of expedition tents produced per day

P = total daily profit

Step 3. Formulate the objective function. This is what you want to maximize or minimize.

The objective is to decide how many of each tent model should be produced each day to maximize profit.

Objective function:

$$\text{Max } P = 500x + 800y$$

Step 4. Formulate the constraints inequalities. These are expressions that limit the amount of resources you can use and the restrictions of your constraint variables.

Cutting department constraints

	Daily cutting time for standard tents		Daily cutting time for expedition tents		Maximum labor-hours per day In cutting department
OR	**1X**	**+**	**2Y**	**≤**	**32**

Assembly department constraints

	Daily cutting time for standard tents		Daily cutting time for expedition tents		Maximum labor-hours per day In assembly department
OR	**3X**	**+**	**4Y**	**≤**	**84**

Non-negativity constraints (since it is not possible to produce negative quantities of tents)

$X \geq 0$ and $Y \geq 0$ **OR** $X, Y \geq 0$

Mathematical Model

Maximize $P = 500X + 800Y$ - *objective function*

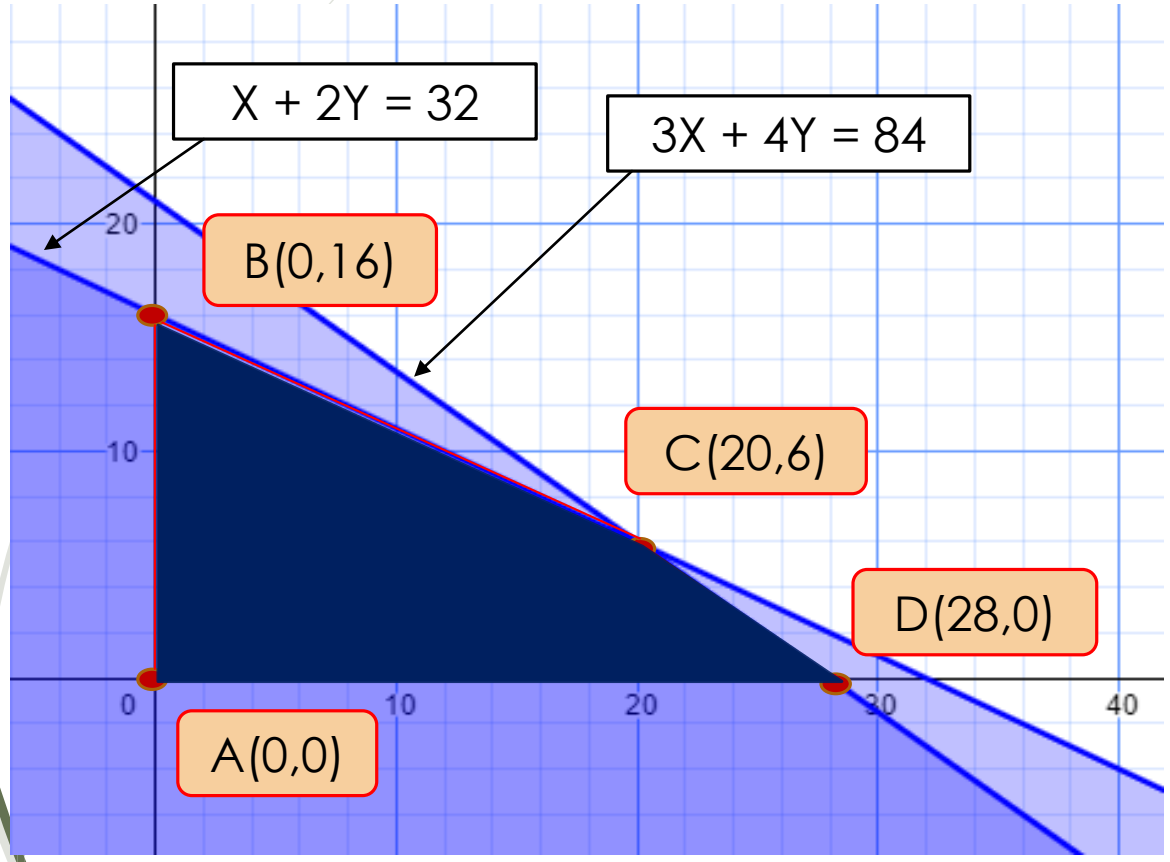
Subject to: $X + 2Y \leq 32$

$3X + 4Y \leq 84$

} *problem constraints*

$X, Y \geq 0$ - *nonnegative constraints*

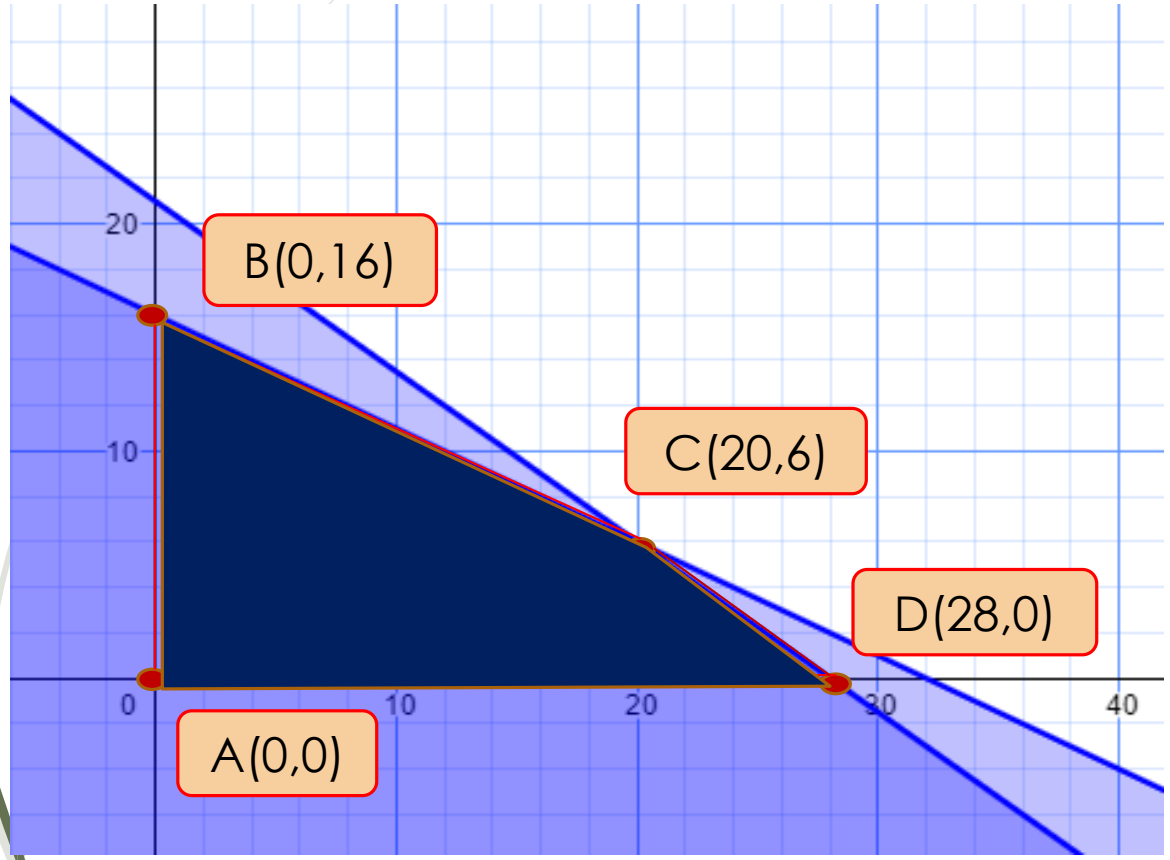
Graphical Method



- The points A,B,C and D are called CORNER POINTS.
- These points (vertices) most likely carry the OPTIMAL SOLUTION to the LP problem.
- The shaded area bounded by the lines is called the FEASIBLE REGION.

Step 5. Solve the problem using an appropriate algorithm or mathematical technique manually.

Max P = 500x + 800y



@ point A (0,0):

P = (500)(0) + (800)(0) = P0

@ point B (0, 16)

P = (500)(0) + (800)(16) = P12,800

@ point C (20,6):

P = (500)(20) + (800)(6) = **P14,800**

@ point D (28,0):

P = (500)(28) + (800)(0) = P14,000

Ergo, The **profit P14,800** is the maximum, hence, the manufacturer must produce **20 standard tents** and **6 expedition tents** daily to maximize profit.

The liquid portion of a diet is to provide at least 300 calories, 36 units of vitamin A and 90 units of vitamin C. A cup of dietary drink X costs $0.12 and provides 60 calories, 12 units of vitamin A, and 10 units of vitamin C. A cup of dietary drink Y costs $0.15 and provides 60 calories, 6 units of vitamin A, and 30 units of vitamin C. How many cups of each drink should be consumed each day to obtain an optimal cost and still meet the daily requirements?

Let x be the number of cups of dietary drink X and let y be the number of cups of dietary drink Y.